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Task Assignment System for Remote Work by Skill Matching: Integration of Web and Mobile Technologies

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Abstract

The spread of coronavirus disease 2019 (COVID-19) has had a significant impact on most companies, requiring them to maintain high levels of operational efficiency. One of the strategies that businesses have chosen is to use technological technology to enable workers to work remotely from home. As a consequence, companies are looking for technologies and software that will enable them to effectively distribute the company tasks to their employees through group work. For such problems, it is a challenge to design and develop an employee assignment system to insist supervisors or department heads to choose right employees using skill matching. Our developed system, entitled Task Assignment System for Employees (TASE) offers a potential technique of task allocation for members based on a skill matching score within remote work environments. We hope that our work, including the implementation of TASE, will inspire further innovation in various fields, allowing researchers to offer innovative ideas for future job assignment systems.

Keywords: Task assignment, skill matching, remote team, mobile application

1. Introduction

Working from home (WFH) is an alternative way providing an opportunity to prevent the spread of COVID-19 by adopting social distancing procedures (Rysavy & Mich, 2020), (Orsini & Rodrigues, 2020). As a result, it leads more chances for employees to work from home. A communication channel is necessary to be built in such a way as to help them collaborate effectively. Working remotely has an impact on communication and cooperation if one-way communication methods, such as email and text messaging, are used more often in the office. This results in various challenges in group work, affecting the company's outcome. Soga *et al*. (2021) indicate that work from home decreases interactions. Furthermore, the lack of face-to-face communication can also lead to misunderstandings of messages causing weak teamwork and low productivity. Therefore, organizations need to establish effective communication tools to ensure that employees working remotely can collaborate effectively. Moreover, working remotely may cause additional issues such as forming uneven groups and balancing the workload of each employee. Communication between team members becomes more difficult when human ties deteriorate, which can lead to a reduction in the time spent asking coworkers for information. This situation can produce an overburdening of any one group or distribute work that does not match a person's talents and abilities, leading to some members of the organization becoming inactive or operating inefficiently. Also, it leads to fewer relationships among coworkers. Consequently, the job is not finished according to the established objectives or on schedule as expected (Smite *et al.*, 2021). The current technology motivates us to develop a system combining a web-based application and a mobile application to aid company supervisors in allocating proper employees to the company's jobs. To achieve this task allocation, it is important to consider both employee's skill and a matching function.

This paper consists of seven sections, including an introduction. Section 2 illustrates related works. Section 3 presents the outline of the matching score method used in our work, involving assessing employees' skills and attributes to match them with job requirements. Examples are also provided to illustrate how the matching score is calculated. The system design and program development are highlighted in Section 4, while Section 5 presents the web and mobile design, including integration. Section 6 provides a comparison in tabular form with other existing research cited in this work to understand the efficiency in task assignment and system performance. The last section discusses our findings and system's limitations, including future research directions.

2. Related Works

In present, task allocation has increasingly employed in most organizes. When a company has to distribute workers for suitable tasks, the manager, who are in charge of the company, has to find the best alternative way to make the best decision. If the number of employees and tasks is large, assigning the right member to the right task will be even more difficult. In such a problem, a task allocation system is required as it can help assign proper workers to tasks efficiently. In most cases, some other factors may be added to the assignment process, such as members' location, utilization of resources, educational background, and soft skills. As a result, task allocation systems can help match employees with the required skill set to tasks, leading to better outcomes and increased productivity. Task allocation systems can help in reducing the time spend to select right employees for the right task. It also helps ensuring a fair distribution of tasks among employees. Furthermore, this can result in increased job satisfaction. Researchers, such as Shehory and Kraus (1995), Verkuilen (2005), and Zhang and Piao (2022)., have been interested in assigning tasks to members in an appropriate manner for a long time. Assigning tasks to members in an appropriate manner implies encouraging others to cooperate to achieve desired results. Group work relies on member cooperation and coordination to extract efficiency in individual work. Many organizations use group work to manage individuals within the organization effectively. Executives or leaders who allocate persons to the relevant group and the right task help make the work most effective while working together. Building good relationships within a group is crucial for successful cooperation in present and future careers. The collective conduct of workers towards each other is essential. According to Miller *et al*. (2021), Covid-19 has transformed the way software development teams cooperate and communicate. Balderas *et al*. (2022) defined a group decision making problem (GDMP) as a participatory procedure where numerous persons participate. Task allocation is another term for group decision making. Scerri *et al*. (2005) proposed a novel distributed task allocation algorithm for large-scale agent teams. Some techniques use the genetic algorithm to optimally allocate members to proper groups (Pedro *et al.* (2002) proposed a distributed algorithm based on a partial centralization technique, called cooperative mediation, to solve distributed constraint optimization problems. Several researchers have proposed a framework for allocating software development work in distributed teams, such as such as Barcus and Montibeller (2008), Amrit (2005), Yin *et al.* (2021), and Shafiq *et al.* (2021), have proposed frameworks for allocating software development work in distributed teams.

The works presented above motivated us to construct a concrete system for task allocation. In addition, engaging in task allocation with the intention of performing optimally is a highly demanding endeavor. However, in this paper, we consider a task assignment problem where multiple individuals with different skills are needed to complete the task. Therefore, a matching score is used to select the most suitable candidates for the given tasks. Skill matching is a method that efficiently aligns many job requirements with employers by maximizing the alignment of required skills (Ren *et al*., 2021). Hence, our system requires job descriptions and personal skill and specifications as input.

3. Matching Scores Used in Employee Allocation

This section presents the concept of a matching score used to evaluate how well an individual. The authors propose a system for allocating tasks to employees based on their skills using a matching score system that considers both primary and secondary skills, with the relative importance of these skills decided by the supervisor.

3.1 Matching Function

The matching score helps a right candidate's worker to the specific task based on the requirement. In common scenario, this can be calculated based on employee’s skills related to the tasks' requirement. Only the skills that are added to the employee's profile have a specific contribution to the matching score. In this work, we match task skills to the employee's skills using the matching score model, which can be calculated by (1). If there are m employees, they are indicated as $n_1, n_2, \ldots, n_m$. In this equation, t_1 and t_2 are determined by the supervisor to prioritize the importance of primary and secondary skills respectively. In this work, we employ 5-point scale ranging from "essential" (scored 5) to "not important at all" (scored 1). For example, if $t_1 = 5$ and $t_2 = 2$, the primary skill is essential, while the secondary skill is approximate 60% less important than the primary skill. In our work, a primary skill is a career of employees which specializes the most, while soft skills are considered secondary skills. However, these values can be added and modified by the supervisor if necessary. An employee who more skills that match the task's requirement, whether it is a primary or secondary skill, will have more chances to be selected by the system. Even if they have a lower matching score. Furthermore, we provide supervisors with access to information for all candidates, enabling them to make better decisions.

$$Score(n_i) = t_1 * (\text{primary skills matched}) + t_2 * (\text{primary skills matched}) \quad (1)$$

, where $1 \leq i \leq m$.

3.2 Example of Matching Function

Suppose there are nine employees, n = 9. A job requires the cooperation of three employees, each of whom possesses specific skills. The required primary skills are "a" and "c", while the secondary skill is "e". Suppose there are nine workers in the company declared their personal skills as listed in Table 1. And, the values of t1 and t2 for calculating score in (1) are 2 and 1, respectively.

TABLE 1: Example of Nine Employees with Personal Skills

Employee No.	Personal skills						
	Primary skills			Secondary skills			
	a	b	c	d	e	f	g
n_1	✓	✓	✓	✓			✓
n_2		✓		✓	✓		
n_3			✓				✓
n_4		✓			✓		✓
n_5			✓		✓	✓	
n_6	✓		✓	✓	✓		
n_7		✓				✓	
n_8	✓			✓		✓	
n_9	✓	✓			✓	✓	✓

Table 2: Total Matching Scores for Nine Employees Based on Criterion (1)

Employee No.	Personal Skills							Total Matching Scores
	Primary Skills			Secondary Skills				
	a	b	c	D	e	F	g	
n_1	2	✓	2	✓			✓	4
n_2		✓		✓	1			1
n_3			2				✓	2
n_4		✓			1		✓	1
n_5			2		1	✓		3
n_6	2		2	✓	1			5
n_7		✓				✓		0
n_8	2			✓		✓		2
n_9	2	✓			1	✓	/	3

Table 3: Sorted Results Based on Total Matching Score, with the Highest Indicating the Best Match.

Employee No.	Personal Skills							Total Matching Scores
	Primary Skills			Secondary Skills				
	a	b	c	d	e	f	g	
n_6	2		2	✓	1			5
n_1	2	✓	2	✓			✓	4
n_5			2		1	✓		3
n_9	2	✓			1	✓	✓	3
n_3			2				✓	2
n_8	2			✓		✓		2
n_2		✓		✓	1			1
n_4		✓			1		✓	1
n_7		✓				✓		0

All members who possess the primary skills in columns a and c of Table 2 are given two points. Additionally, give one point to everyone who possesses the secondary skills. The outcomes of each member's scores are shown in Table 2 in which n_6 is the best with the core of 5. A. We then reorder everyone according to their final score. Table 3 illustrates the outcome. For the last required skill, 'e', we found 5 employees: n_6, n_5, n_9, n_2, and n_4. As a result, there are 4*4*5 = 80 possible combinations the specific task. For clarification and better understanding, the number shown in Figure 1 corresponds to the employee number; for example, 1 signifies employee n_1. Based on our searching method for possible combinations, it is possible that some individuals can be available more than once within the same group. This kind of combination is considered as an unqualified group. For instance, the combination of 666 has only n_6 as the member of the group. We then need to remove it from the list. We demonstrate the possible combinations of this example in Figure 1.

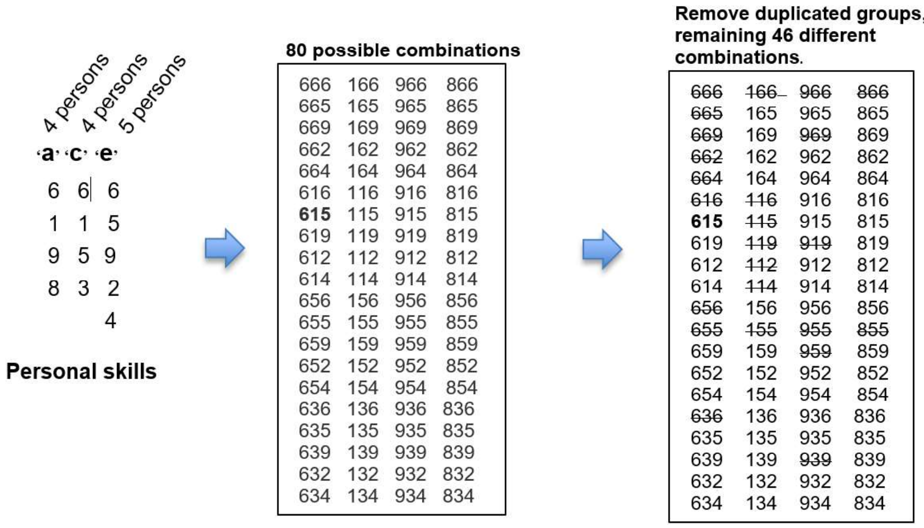


Fig. 1: Possible Combinations Based on Personal Skills

Furthermore, some combinations may appear more than once in all possible combinations. For instance, 659 and 956, they imply the same group of n_5, n_6, and n_9. Hence, after removing these duplicated groups, there are only 46 different ways. We then sort all group by its score calculated by (1) in order to determine the best one. The total score of this group becomes score(n_6) + score(n_1) + score(n_5) = 5+4+3 = 12. For this example, a group of n_6, n_1, and n_5 represented as 615 is designated as the recommended group.

4. System Design and Program Development Process

In this work, two different platforms have been developed using Xcode and Visual Studio Code. Because remote work environments need clear and efficient task management, TASE is designed to include two parts based on user type. The first part helps supervisors assign tasks effectively, while the second part allows employees to easily access their assignments.

As the TASE developed within remote work environments, The Tasks Assignment System for Employees is divided into two parts based on user type:

1. Supervisor: This web-based application is indeed designed for supervisors or organizational heads who need to assign tasks to their team members.
2. Employees: An iOS mobile application specifically designed for employees, enabling them to access and manage their assigned tasks efficiently.

Figure 2 depicts the system's architecture, providing a broad overview. In our designed system, Firebase Cloud Messaging (FCM) is used as a cross-platform messaging system to send notifications to employees on the iOS application. This ensures that important updates, such as new task assignments or changes to existing tasks, are promptly communicated. Additionally, the system leverages FCM's ability to handle various types of data messages, ensuring that employees receive detailed information directly on their devices. This integration is important for keeping communication smooth in remote work settings, ensuring updates are on time and information is shared effectively.

Figure 3 illustrates the 11 use cases of our system. The first seven use cases are tailored for supervisors, utilizing the web interface and leveraging Firebase for data storage. The mobile app platform equipped with a real-time database and unified client libraries across various mobile programming languages, enables efficient data management. The remaining use cases are designed specifically for employees, providing functionalities such as task assignment based on skill sets, notification management via Firebase Cloud Messaging, and ensuring seamless communication and operational efficiency within remote work environments.

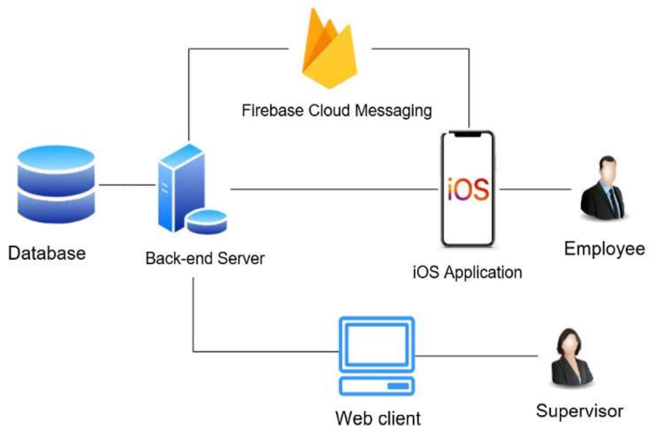


Fig. 2: Architecture of the system

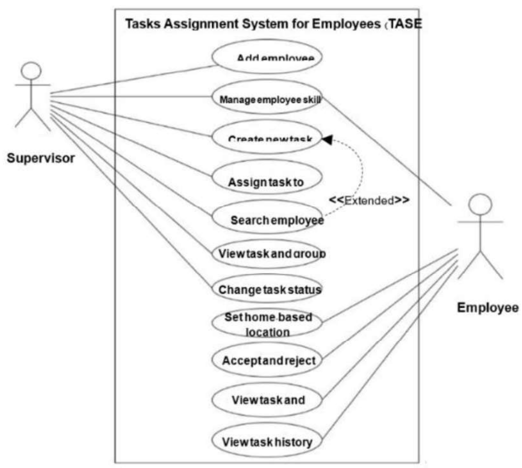


Fig. 3: Use case diagram for TASE

4.1 Supervisor on Window-Based Application

The first seven use cases are for the supervisor and run on the web while storing data in Firebase. Therefore, everything will be instantly synced with the mobile app. One reason for building the web-based supervisor part is that managing duties and workers is quite detailed. Working on the web allows users to access all information and be more adaptable. The remaining use cases are intended for employees and can be accessed through a mobile application on iOS.

1. Add Employee: Supervisors can add employees' information such as first name, last name, user type, email, mobile phone number, core competence, and auxiliary capabilities.
2. Manage Skill and Information: After the employee has been added to the system, they can manage their own abilities, and the supervisor can also modify them.
3. Create New Task: Supervisors can establish a new task with job descriptions such as title, number of employees, start and finish dates, and key skills required for the position.

4. Manage Employee Skill and Information: Users can input information to establish primary skills, secondary skills, and required fields to generate recommended lists of groups.
5. Search Employee: The system allows for searching individuals with specific skills using the previously described matching function.
6. View Task and Group: Supervisors can track the overall performance of the group and the progress of tasks allocated to employees.
7. Change Task Status: The supervisor can modify the work status, such as “Cancel” if the job is canceled, or “Late operation” if the due date has passed. If the job is already assigned to employees, its status becomes “In operation”.

4.2 Employee on Mobile Application

1. Set home-based location: Setting a "homepage location" is necessary for group members to see the distance to each member or to arrange on-site group meetings. This helps them understand the members' locations on a map and plan accordingly if necessary.
2. Accept and reject assigned jobs: This use case demonstrates how the system prioritizes both managers and employees. Employees who receive a notification from their employer can decide whether or not to accept the job. If they decide to reject it is required to fill in the reason saved in the system for future assignments as well.
3. View task and team member: Workers may access the information of assigned tasks, such as the job title, task schedule, start and finish dates, and a list of employees who have consented to work as assigned.
4. View task history record: Employee work history is a collection of prior employment, including current occupations, that employees may examine to understand their current situation. This historical record is also useful for employees when accepting new assigned jobs since it informs them of the current job number.

4.3 System Constraint and Limitations

System constraints help us understanding a roadmap for future development, including the system's limitations. Hence, developers can identify areas that need improvement for the future. The limitations of our system are:

1. No support for subtasks: The system only supports normal tasks.
2. No personnel hierarchy support: The system only supports a single level of management.
3. No evaluation of individual and groups’ performance.
4. No dynamic location for employees involved in the task assignment.

5. User Interface Design

This section presents the interface design for two groups: supervisors and workers. Each part is described in detail as follows.

5.1 Supervisor Part:

The Figures 4-8 are screenshots of the web application designed for the supervisor. As mentioned earlier, the system will not consider jobs consisting of multiple subtasks. However, when supervisors start a new job, they must create a job with essential descriptions, including the number of people required to complete the job, the essential skills that each employee needs to have, and other personal soft skills (see Figure 4 for details). Additionally, if all jobs require the same skill, the manager may check the box labeled "all qualities are the same" to make the process of choosing employees easier. If not, the manager can add the required skills for the individual job. The skills are categorized into two groups: primary skills and secondary skills. An example screenshot of choosing both primary and secondary skills is presented in Figure 5. After the new job is generated, it automatically searches for available employees that match the job's requirements. Setting conditions for selecting suitable employees might include choosing from the employee's primary skills as well as secondary skills.

Figure 6 illustrates a screenshot of the result of grouping based on matching score. The best group, marked as the "recommended group," will be presented at the top of the other groups. Also, the number of employees' jobs in hand will be reported to the supervisor for the correct decision to be made. In practice, the supervisor will select employees who have not yet been assigned to any work. After an employee is assigned a specify task, he or she may accept or decline the position for personal reasons, such as an overburdened workload. If the required skills map to anyone, that employee will be selected to present to the supervisor. The supervisor can select the best one, as shown in the "Results of recommended group". The recommended groups can be generated more than one, if they have the same matching score according to (1). In addition, the groups with lower matching scores will be displayed below for the supervisor to reconsider the decision if the recommended groups are not satisfactory. This make the choice for the supervisor to choose. Moreover, members can be added or removed at any time. Once the group has been set, the supervisor may see and manage each employee's information on the company list, as shown in Figure 7. Figure 8 depicts how the task status is changed by the supervisors. The job can be assigned one of the following statuses: "in operation", "late operation", "success", or "cancel". If the job is still active, its status is "in operation". If the job is not running as it is cancelled by an authorized manager, it can be marked as "cancel". When the job is completed, its status is "success". However, if the job surpasses the due date, it is marked as "late operation". The term "late operation" refers to work that is still ongoing, but has passed its deadline. “Success” is assigned to the completed task. The cancelled job will be noted as “cancel”. However, if the work reaches the state with the appropriate status, it can be completed automatically.

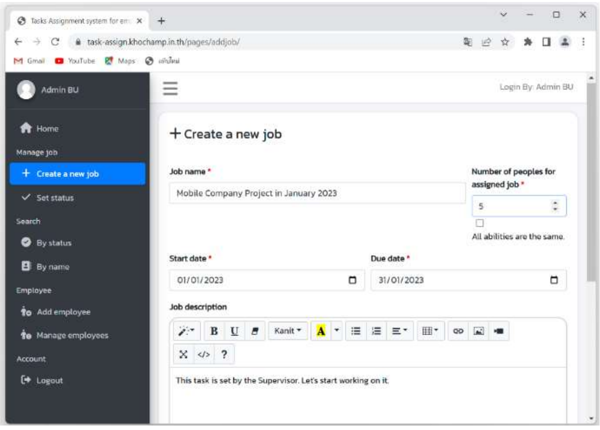


Fig. 4: A screenshot for “Create a new job”

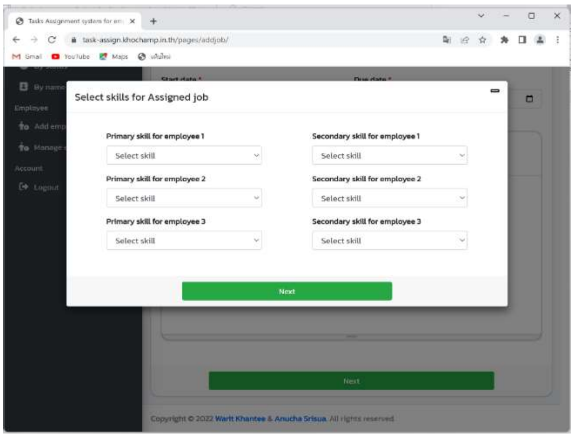


Fig. 5: An example screenshot of selecting specific skills for new job.

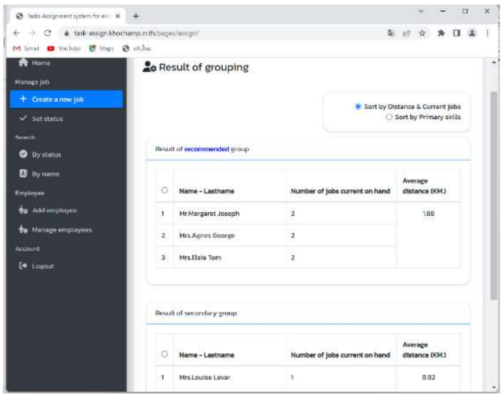


Fig. 6: Screenshot of results from grouping based on matching score

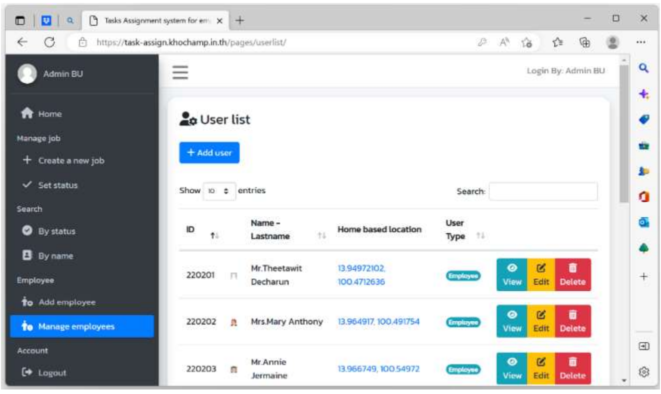


Fig. 7: Screenshot of current user lists added by the supervisor.

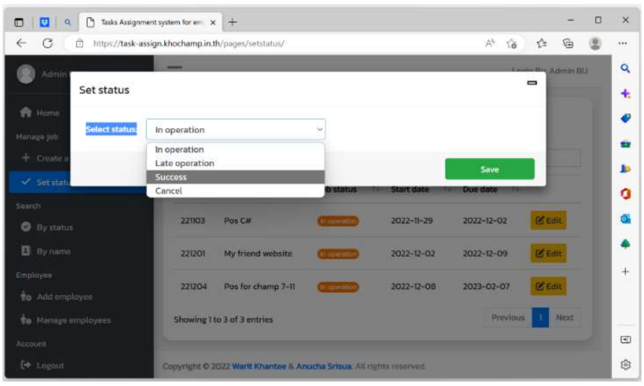
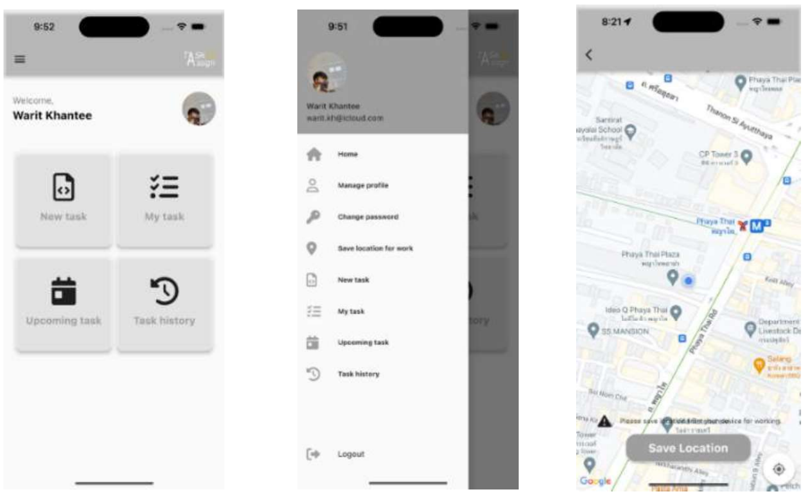


Fig. 8: An example screenshot of setting job's status

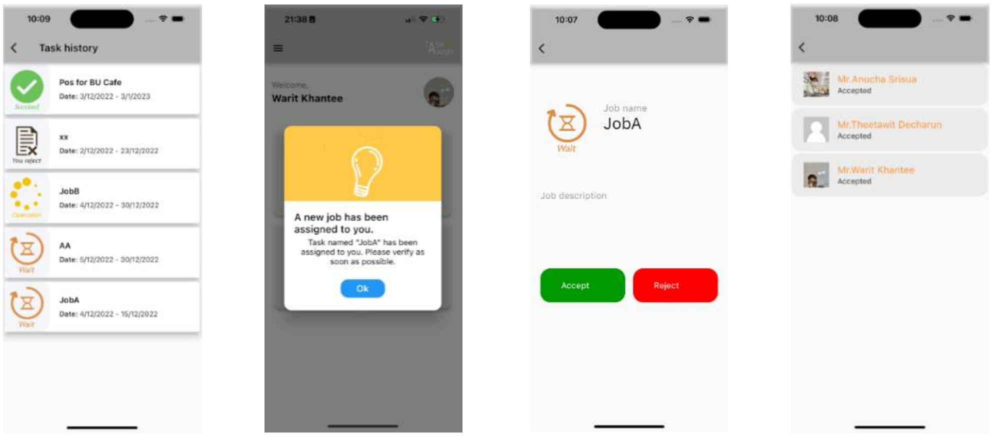
5.2 Employee Part:

Figures 9 and 10 demonstrate screenshots of a mobile application developed for employees. Figure 9(a) depicts what appears on an employee's mobile device after logging into the system. Options such as “New task,” “My task,” “Upcoming task,” and “Task history” can be selected. The employee's name and picture appear in the upper part of the program. When the user clicks on ≡, the menu will shift slightly to the left of the screen, as seen in Figure 9(b). Figure 9(c) shows how to configure the home-based location on the map. The history of all jobs that can be taken can be seen in Figure 10(a). If a new task is assigned to the user, the notification will appear both in the mobile application and by email. An example screenshot of the mobile app is presented in Figure 10(b). As previously stated, the assigned employee is free to accept or decline the assigned assignment. Figure 10(c) shows a screenshot of the employee’s approval of the new assigned task. Click the “Accept” button if the user accepts the job; otherwise, click the “Reject” option. If all members have been verified to take the assigned job, all members may browse their team members and read work for that assignment, as shown in Figure 10(d).



a) Main section of the app b) Main menu c) Home-based location

Fig. 9: Screenshot of the mobile app



a) Task history b) New job notification c) Approve of job assignment d) Peers in the assigned group

Fig. 10: Example screenshot related to user's task assignment

6. Comparison of Task Assignment Methods and Discussion

To highlight the contributions of TASE, we compare previous studies with others in the related field. This includes examining the works of Verkuilen (2005) utilizes fuzzy logic for task matching and Zhang and Piao's (2022) management of complex multi-agent tasks. Furthermore, Balderas et al. (2022) emphasis on decision-making speed. As shown in Table 4, TASE focuses specifically on optimizing task assignments within business contexts, aligning individual skills with organizational requirements. Based on Table 4, Verkuilen utilizes fuzzy set analysis to assign membership levels, aiming to match employees across a spectrum of suitability, offering flexibility but potentially lacking the precision of skill-based matching. Zhang and Piao (2022) utilize Vein-Based Multi-Agent Pattern Formation (VB-MAPF) to coordinate agents in complex task scenarios, tailored for intricate multi-agent systems. Balderas *et al.* (2022), on the other hand, focus on multi-criteria ordinal classification for group decision-making, prioritizing accuracy and speed in consensus-building within group settings. In contrast, TASE employs a skill matching score approach to allocate tasks, which emphasizes optimizing based on individual skills. This method is particularly effective in improving task allocation time for simple or plain tasks. It does not focus on or support complex businesses with hierarchical subtasks, which may need more advanced and customized approaches to manage tasks.

Table 4: Comparison of Task Assignment Systems

Aspect	TASE	Verkuilen (2005)	Zhang and Piao (2022)	Balderas *et al.* (2022)
Methodology	Skill matching score for task allocation	Fuzzy set analysis for assigning membership levels	Vein-Based Multi-Agent Pattern Formation (VB-MAPF)	Multi-criteria ordinal classification for group decision-making
Focus	Optimizing task assignments based on individual skills	Matching employees based on suitability spectrum	Coordination of numerous agents for complex task formations	Optimizing consensus in group decisions
System Type	Business task assignment	Fuzzy logic-based assignment	Multi-agent coordination	Group decision support
Performance Metrics	Task allocation time, productivity improvement	Accuracy of membership level assignments	Success in pattern formations	Consensus accuracy and speed
User Interface	Web-based for supervisors, mobile for employees	Not specified	Not specified	Interface for group decision support
Applications	Business needing efficient task assignment	Complex employee task matching	Complex multi-agent tasks	Group decision environments

The main focus of TASE is on employees' skill and matching function for task allocation; therefore, TASE may not be as versatile or adaptable in complex organizations. It becomes the weak point of TASE, as it does not focus on complex task formations and handle the coordination of numerous agents as effectively as seen in the Vein-Based Multi-Agent Pattern Formation (VB-MAPF) method of Zhang and Piao (2022). In contrast, Verkuilen (2005) relies on expertise in fuzzy logic, while Zhang and Piao require proficiency in multi-agent coordination, both adding complexity to their implementation and potentially limiting their applicability without specialized knowledge. Also, Balderas *et al.* (2022) prioritizes achieving accurate consensus in group decision-making, potentially causing longer delays in task allocation and requiring a more complex implementation process. Based on Table 5, TASE stands out in other aspects: optimizing task allocation time, scalability across business sizes, and ease of implementation. TASE assigns tasks based on employees' skills using the matching skill function, and the focus is on distributing the company tasks to employees based on the business environment and employees' skills. With the design of the TASE's matching function, it is able to ensure quick and accurate management of the task allocation process and straightforward implementation. Therefore, TASE's scalability highlight its advantage in enhancing operational efficiency across different organizational contexts.

Table 5: System Performance Aspects

Performance Aspect	TASE	Verkuilen (2005)	Zhang and Piao (2022)	Balderas *et al.* (2022)
Task allocation Time	Based on the efficiency of skill matching function	Variable based on fuzzy set analysis	Varies with complexity of pattern formation	Depends on the speed of consensus decision-making
Scalability	Scalable for businesses of various sizes	Scales with complexity of fuzzy set analysis	Effective for large, complex multi-agent systems	Scales with the size of decision-making groups
Ease of Implementation	Easy to implement with straightforward skill matching	Requires expertise in fuzzy logic	Complex implementation for coordination	Moderate, with need for multi-criteria decision support

7. Concussions and Future Work

TASE is designed for task allocation within remote work environments. The web-based supervisor interface enables team formation using our innovative matching score model, while the iOS mobile app empowers employees with access to task details. In the future, we will investigate modifying the system to enhance its performance and align with the diverse business. Individual and group performance will be considered to leverage the skill matching function. Moreover, we will continue improving the system by adjusting the matching skill function to accommodate to the complexity of the relationships between task with subtasks. We also plan to expand TASE's compatibility to include other operating systems and cross-platform compatibility, making it accessible to a wider range of users. We will explore the use of machine learning algorithms to improve the level of user satisfaction and accuracy of the matching scores by continuously learning from new data. Furthermore, we will utilize AI to ensure that the system efficiently matches the best individuals with the relevant tasks. This improvement aims to ensure that each task is assigned to the best-suited candidate based on their abilities and qualifications.

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